

Three-Dimensional Multiple Object Tracking Speed Thresholds are Associated with Measures of Simulated Driving Performance in Older Drivers

Heather Woods-Fry¹, Swanti Deut¹, Charles A. Collin¹, Sylvain Gagnon¹, Jocelyn Faubert³, Michel Bédard⁴, Shawn Marshall⁵

¹Department of Psychology, University of Ottawa, Ottawa, Ontario, Canada

²Transportation Safety Board of Canada, Ottawa, Ontario

³Visual Psychophysics and Perception Laboratory, Université de Montréal, Montréal, Québec, Canada

⁴Centre for Research on Safe Driving, Lakehead University, Thunder Bay, Ontario

⁵Ottawa Hospital Research Institute, Ottawa, Ontario

We assessed the relationship between three-dimensional multiple-object-tracking (3D-MOT) ability and older driver simulated driving performance. Participants included 30 older drivers from the local community who completed two experimental driving scenarios, as well as a three-dimensional multiple object tracking task, called the 3D-MOT task. The speed thresholds on the 3D-MOT task were strongly negatively associated with lane deviation and crash rates during a simulated highway drive. Our findings help to expand the existing knowledge regarding multiple object tracking and driving, as it is commonly assumed that some aspects of driving include the ability to track the movement of many objects. Our findings further elucidate the relationship that motion processing has with driving performance in the older driver population.

INTRODUCTION

When examining the type of visual information that we encounter in our daily lives, a large amount of what we perceive is in motion. Not only must we be able to effectively process the visual motion in our environment, but we must also be able to accurately track the moving objects of relevance so that we can know their current location in relation to us and anticipate their subsequent movements. A task known as multiple object tracking (MOT), introduced by Pylyshyn and Storm (1988), is assumed to be widely applicable to such every day tracking tasks, and is considered to be a laboratory-based analogue for real world tracking. The traditional MOT task assesses the ability to track the position of a number of target items as they move among featurally identical distractors.

Many researchers assert that multiple object tracking is vital to driving (Fencsik, Klieger & Horowitz, 2007; Fera, 2008; Horowitz et al. 2007), however only a couple of studies have directly examined the relationship between MOT and driving (Bowers et al., 2013; Lochner & Trick, 2014). Bowers et al. (2013) found that MOT results are predictive of the on-road driving performance of older drivers, such that older drivers with poorer results on the MOT task were at risk of committing highway, observation, planning, speed control, and indication errors. Lochner and Trick (2014) asked the vital question of whether multiple object tracking could even be performed while driving. Therefore, they formulated several experiments that investigated if there was any interference from performing multiple object tracking in the context of a driving task. Their experiments consisted of a multiple object tracking task that was adapted to be performed in a simulated driving environment. In the first experiment, the participants completed a driving scenario while having to keep track of a set of moving vehicles in an array of other vehicles on the

road. They found that as the participants had to track more objects (1, 3 or 4), their ability to maintain a consistent headway while driving was compromised. Average deviation in headway maintenance was significantly larger when tracking four targets in comparison to one or three targets. Deviations in lateral positioning also followed the same trend. A second experiment examined whether tracking while driving would enhance the drivers' ability to respond to changes in the tracked vehicles. They found that participants were more successful at detecting featural changes in tracked versus untracked vehicles, in addition to making faster detections when a tracked vehicle swapped properties with another vehicle. Most importantly, they found that drivers were 14% more accurate at detecting when a target vehicle braked as opposed to a distractor vehicle.

Both aforementioned studies examined important aspects of driving performance in relation to multiple object tracking ability. The current study was designed to be exploratory in nature, and expand upon the small amount of research in this area. The current study examined how a measure of multiple object tracking, called the three-dimensional multiple object tracking (3D-MOT) task, is related to the driving ability of older adults. The latter was measured via two experimental scenarios in a mid-fidelity driving simulator. Using a driving simulator allowed us to expose our drivers to simulated hazardous events that would not be ethical to test during an on-road assessment and thus assess the relationship between the 3D-MOT and driving.

The 3D-MOT was created by Legault, Allard and Faubert (2013). Instead of assessing number of items successfully tracked, as is common in other MOT tasks, it uses object speed as the dependent variable. Legault et al. (2013) chose this dependent variable because it can vary on a continuous ratio scale and is not affected by the number of objects that can be tracked at once. Because older adults can

usually track around 3 objects (Trick, Perl & Sethi, 2005), this is the number we used in our version of the 3D-MOT task. That is, instead of focusing on the number of items successfully tracked, this version of the MOT uses three targets at all times and instead assesses the individual differences in one's capabilities to track the prescribed number of targets in terms of threshold speed of motion (Faubert & Sidebottom, 2012).

The 3D-MOT task used here also varies from the traditional task in that it is presented three dimensionally. It has been demonstrated in the literature that three-dimensional multiple object tracking differs from the 2D variety. For instance, it produces better speed threshold performance, likely due to the higher ecological validity (Tinjust, Allard, and Faubert, 2008). Tinjust et al. (2008) demonstrated that multiple object tracking speed thresholds were significantly better when the task was presented in stereoscopic versus non-stereoscopic conditions. Another study found that occlusion cues and binocular disparity enhanced the number of items successfully tracked (Viswanathan & Mingolla, 2002).

Although there are only a handful of studies that have explored how multiple object tracking is related to driving, we hypothesize that speed thresholds gathered with the 3D-MOT task will be negatively associated with relevant driving measures. Both experimental driving scenarios are well suited to being able to tap into situations where drivers would have to track multiple road events. Therefore, we hypothesize that 3D-MOT thresholds will be negatively correlated with older drivers' simulated crash rate, as well as lane deviations.

METHOD

Participants

Older drivers were recruited using advertisements in the community. Participants were 30 volunteers with ages ranging from 65-85 years old ($M = 72.5$, $SD = 5.7$). Participants had no existing eye disease. They were required to possess a G2 license for a minimum of 5 years prior to testing. The participants received 30\$ compensation for their time.

Procedure

Upon reviewing and signing the consent form, participants were administered the Montreal Cognitive Assessment (MoCA) (Nasreddine et al., 2005) ($M = 25.3$, $SD = 2.44$). After this, participants filled out the Simulator Sickness Questionnaire (SSQ) (Kennedy et al., 1993) to gather a baseline to compare to any changes in symptoms due to simulator sickness subsequently produced by the simulated drives. Participants drove three simulator scenarios, including two experimental scenarios whose order was counterbalanced. The SSQ was filled out at the end of each simulator scenario in order to monitor any changes in symptoms. Participants then completed the multiple object tracking task.

Out of 30 participants, only 11 were able to complete all simulator scenarios due to simulator sickness. The individual scenarios had a higher completion rate. There were 15 participants who completed the city driving experimental scenario and 16 participants who completed the highway

merging experimental scenario. All incompletions were due to symptoms of simulator sickness

Three-Dimensional Multiple Object Tracking (3D-MOT)

The 3D-MOT task is a technique that is used to study how the visual system tracks moving targets in three dimensions. Participants were asked to track three spherical targets moving amongst four featurally identical distractors. That is, a total of seven yellow spheres were presented. At the start of each trial these were static. After 2 seconds, the three target spheres turned red with a surrounding white halo. This highlighting remained for 2 seconds, after which the targets turned back to yellow. The spheres then started to move for 8 seconds along a linear path within the constraints of the screen (i.e., they bounced off the walls of the virtual space they were in). They moved through one another and at times occluded one another. After this, all spheres stopped moving and were labeled with numbers 1-7. Subjects used the mouse to click on the three spheres they believed to be the targets. Once their response was locked in, the correct target spheres were revealed and feedback was given. After each trial, the speed of the sphere movement for the next trial was increased if the subjects' answer was correct, or else it was reduced. The independent variable was the speed threshold for tracking three targets. The test consisted of 20 trials and the threshold was estimated using the staircase method. The duration of the test was approximately 10 minutes (Legault et al., 2013).

Driving Assessment

A mid fidelity STISIM 3.0 driving simulator by Systems Technology Inc. (systemstech.com; Hawthorne, CA) was used to generate and present the virtual driving environment. Simulations were projected onto three screens (side panels 123cm x 91cm; center panel 138cm x 91cm) that were located 160 cm away from the driver's seat, providing a 180° forward field of view. A modified Logitech G25 steering wheel was used along with pedals from Extreme Competition Controls Inc (ECCI). The virtual environment was coordinated with realistic audio effects allowing for acceleration cues. The STISIM 3.02.13 Drive software collects several variables, including vehicle position and motion, as well as the status of the various driving controls, at a rate of 10 Hz. The software was run on a Windows XP operating system with four Dell Dimension 9200 computers (2.4 GHz Intel Core 2 Duo processor, 3 GB RAM per computer).

Participants completed three simulator scenarios. To begin, all participants completed a familiarization scenario. Two experimental driving scenarios were then completed: a city driving scenario and a highway merging scenario. The order of the experimental driving scenarios was counterbalanced.

In the familiarization scenario, participants drove on a two-lane road for 5.7 km where they learned how to maintain their speed, perform a complete stop and navigate left and right turns. The city driving scenario was 7.8 km long. The setting began in a downtown environment, where participants drove through a small town, followed by a rural road and then

a city environment. A total of 5 unexpected events were presented during this scenario: 1) on a 6-lane road, a car merged onto the road from a gas station on the right; 2) on a 6-lane road, a car ran a red light from the left just as the participant was crossing an intersection; 3) on a 2-lane road, a car merged onto the road from the shoulder; 4) on a 2-lane road, a moose ran across the road from the right; and 5) on a 4-lane road, a car merged onto the road from a right-side merging lane. The highway merging scenario was a 10.7 km drive that required participants to enter and exit a highway numerous times. Participants were verbally prompted by the simulator to perform lane changes to the left-most or right-most lane. As they advanced in the scenario, traffic levels increased, thus making it more difficult to perform the instructed lane changes.

During the driving scenarios, our variables of interest were recorded automatically by the simulator (i.e., crash rate and lane deviation). Due to the exploratory nature of this research, only broad measures of simulated driving performance were considered. Crash rate was determined by the total number of collisions with objects in the simulated environment. These included, but were not limited to, the objects involved in the 5 unexpected events. We also examined participants' total lane deviation time. This was recorded for the duration of the simulation and was considered as the percentage of time spent outside of a prescribed lane. The dependent variables were crash rate and lane deviation.

RESULTS

3D-MOT & Simulated Driving

To determine whether 3D-MOT threshold scores are associated with simulated driving performance, we calculated the correlation between tracking speed thresholds and simulator measures (crash rate, lane deviation) for both scenarios and each scenario individually. The correlations (one-tailed) can be found in Table 1. A post hoc power analysis was completed for the smallest sample ($N=11$), and it was determined that the analysis was likely underpowered, with a value of .50, indicating that there was a high likelihood of a type II error. Despite this, we find highly significant correlations between 3D-MOT thresholds and both crash rate and lane deviation in the highway driving scenarios. This is not the case with the City Driving scenarios. Putting all scenarios together yields moderate correlations that approach significance.

Table 1
Correlations (one-tailed) between the 3D-MOT and simulated driving measures in older drivers

	Total ($N=11$)	City Driving ($N=15$)	Highway Merging ($N=16$)
Crash rate	-.50 $p=.059$	-.23 $p=.20$	-.58 $p=.009$
Lane deviation	-.52 $p=.051$	-.04 $p=.44$	-.56 $p=.02$

DISCUSSION

We hypothesized that 3D-MOT tracking speed thresholds would be negatively associated with crash rates during simulated driving. That is, we expected that the worse a driver's performance at the 3D-MOT, the more often they would exhibit simulated driving errors. Our results showed that overall, the association between 3D-MOT and total crash rate was trending, but did not quite reach significance, possibly due to high subject attrition and a resultant small sample. However, the individual scenarios differed markedly, demonstrating that 3D-MOT was significantly negatively associated with crash in the highway merging scenario but not in the city driving scenario.

We further hypothesized that the 3D-MOT tracking speed thresholds would be negatively associated with lane deviation during simulated driving. Our results showed that the relationship between the 3D-MOT and total lane deviation almost reached significance. As with crash rates, the results from the two scenarios were quite different. 3D-MOT was highly negatively associated with lane deviation in the highway merging scenario but not the city driving scenario. Thus, our results showed a trend towards individuals with higher tracking speed thresholds being less likely to crash. Furthermore, a similar trend appeared where higher tracking speed thresholds indicated better overall lane maintenance.

A potential explanation for the differences in outcome between the scenarios may come from work by Wickens (2002). He states that driving can be divided into ambient and focal attentional tasks. Ambient attention is used for sensing direction and orientation; it is a more broad and unfocused general sense of one's position relative to one's surroundings. Focal attention is used for detailed tasks, such as reading, pattern recognition, or obstacle detection, essentially anything that requires foveation. Wickens describes lane keeping and general positioning as tasks that rely on the ambient attentional stream, whereas obstacle avoidance is mainly a task of focal attention. It is important to note that the ambient and focal channels are linked, such that the ambient channel provides positional information to the focal channel and thus acts as the foundation for focal attention (Previc, 1998; Oksama & Hyona 2016).

Both ambient and focal attentional channels are thought to be involved in object tracking, depending on the nature of the tracking task. Specifically, the multiple object tracking (MOT) task, where one must know the location of the tracked objects, calls primarily upon the ambient stream; Conversely, the task of multiple identity tracking (MIT) calls upon both the ambient and the focal streams (Oksama & Hyona, 2016). Multiple identity tracking is similar to MOT, except that the identities of the tracked objects are also relevant information, not just the positions of the objects (Oksama & Hyona, 2004). The additional information gathered about the identity of the tracked objects implicates the focal attention stream, because the objects must be foveated in order to be identified (Oksama & Hyona, 2016). The real world task of driving is arguably most similar to MIT, since the driver is aware of their position relative to the surrounding road environment, but also aware of the identities of the different types of road users that are nearby. We had our participants complete a tracking task that is thought to implicate primarily the ambient attentional

channel; this may be why we see a relationship between results on this task and lane keeping (i.e., both are primarily tasks of ambient attention).

The relationship between MOT—which is primarily a task of ambient attention—and crash—which is thought to be a failure of focal attention—is more difficult to explain in the context of Wickens's (2002) model. However, it might be explained based on the idea that the ambient attentional stream acts as the foundation of focal attention. That is, ambient attention gathers general position and directional heading information, which subsequently informs the focal attention mechanism used to identify possible obstacles.

Overall, the current study has helped to expand the existing knowledge regarding multiple object tracking and driving. With further research, multiple object tracking could eventually be considered as a candidate for inclusion in an assessment battery for older drivers. As a cognitive-perceptual task, the MOT is in line with the Multi-factorial Model of Driving Safety by Anstey et al. (2005), and research by Wood et al. (2008), which supports the assessment of visual, cognitive and motor capacities in order to assess older drivers' fitness to drive. Within these three functional domains, both visual and cognitive tests have most often been shown to be associated with older driver behavior (Ball et al. 2006, Vanlaar et al. 2014). Ultimately, this research has served to expand the existing knowledge regarding multiple object tracking and driving, helping to further elucidate the relationship that object tracking has with driving performance in the older driver population.

REFERENCES

- Anstey, K. J., Wood, J., Lord, S., & Walker, J. G. (2005). Cognitive, sensory and physical factors enabling driving safety in older adults. *Clinical Psychology Review*, 25(1), 45-65.
- Bowers, A. R., Anastasio, R. J., Sheldon, S. S., O'Connor, M. G., Hollis, A. M., Howe, P. D., & Horowitz, T. D. (2013). Can we improve clinical prediction of at-risk older drivers? *Accident Analysis and Prevention*, 59(1), 537-547.
- Fencsik, D., Klieger, S., & Horowitz, T. (2007). The role of location and motion information in the tracking and recovery of moving objects. *Perception & Psychophysics*, 69(4), 567-577.
- Feria, C. S. (2008). The distribution of attention within objects in multiple-object scenes: Prioritization by spatial probabilities and a center bias. *Perception & Psychophysics*, 70(7), 1185-1196.
- Feria, C. S. (2013). Speed has an effect on multiple-object tracking independently of the number of close encounters between targets and distractors. *Attention, Perception, & Psychophysics*, 75(1), 53-67.
- Faubert, J., & Sidebottom, L. (2012). Perceptual-cognitive training of athletes. *Journal of Clinical Sport Psychology*, 6(1), 85.
- Horowitz, T., Klieger, S., Fencsik, D., Yang, K., Alvarez, G., & Wolfe, J. (2007). Tracking unique objects. *Perception & Psychophysics*, 69(2), 172-184.
- Legault, I., Allard, R., & Faubert, J. (2013). Healthy older observers show equivalent perceptual-cognitive training benefits to young adults for multiple object tracking. *Frontiers in psychology*, 4, 323.
- Lochner, M. J., & Trick, L. M. (2014). Multiple-object tracking while driving: the multiple-vehicle tracking task. *Attention, Perception, & Psychophysics*, 76(8), 2326-2345.
- Nasreddine, Z. S., Phillips, N. A., Bédirian, V., Charbonneau, S., Whitehead, V., Collin, I., Cummings, J.L. & Chertkow, H. (2005). The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment. *Journal of the American Geriatrics Society*, 53(4), 695-699.
- Oksama, L., & Hyönä, J. (2004). Is multiple object tracking carried out automatically by an early vision mechanism independent of higher-order cognition? An individual difference approach. *Visual cognition*, 11(5), 631-671.
- Oksama, L., & Hyönä, J. (2016). Position tracking and identity tracking are separate systems: Evidence from eye movements. *Cognition*, 146, 393-409.
- Previc, F. H. (1998). The neuropsychology of 3-D space. *Psychological bulletin*, 124(2), 123.
- Pylyshyn, Z. W. (2009). Perception, representation, and the world: The FINST that binds. *Computation, cognition, and Pylyshyn*, 3-48.
- Pylyshyn, Z. W., & Storm, R. W. (1988). Tracking multiple independent targets: Evidence for a parallel tracking mechanism. *Spatial vision*, 3(3), 179-197.
- Tinjust, D., Allard, R., & Faubert, J. (2008). Impact of stereoscopic vision and 3D representation of visual space on multiple object tracking performance. *Journal of Vision*, 8(6), 509-509.
- Trick, L. M., Enns, J. T., Mills, J., & Vavrik, J. (2004). Paying attention behind the wheel: A framework for studying the role of selective attention in driving. *Theoretical Issues in Ergonomic Science*, 5(5), 385-424.
- Trick, L. M., Perl, T., and Sethi, N. (2005). Age-related differences in multiple-object tracking. *Journal of Gerontology Series B Psychological Sciences and Social Sciences*, 60, 102-105. doi:10.1093/geronb/60.2.P102
- Vanlaar, W., McKiernan, A., McAteer, H., Robertson, R., Mayhew, D., Carr, D., ... & Holmes, E. (2014, September). Cognitive screening tools for predicting unsafe driving behavior among senior drivers: a meta-analysis. In *Proceedings of the HFES Annual Meeting* (Vol. 58, No. 1, pp. 2048-2052). Sage CA: Los Angeles, CA: Sage Publications.
- Viswanathan, L., & Mingolla, E. (2002). Dynamics of attention in depth: Evidence from multi-element tracking. *Perception*, 31(12), 1415-1437.
- Weber, B., & Ni, R. (2015). MOT Capacity is Compromised in Ecological 3D Environments. *Journal of Vision*, 15(12), 464-464.
- Wickens, C. D. (2002). Multiple resources and performance prediction. *Theoretical Issues in Ergonomics Science*, 3(2), 159-177.